

Graphical representation of the visual aberrations of biocular magnifiers

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ABSTRACT

A biocular magnifier is an optic that is sufficiently large to be used by both eyes together, and which presents a magnified virtual image at a finite distance from an observer. The design of such an optic is one of the most difficult tasks in optical design due to the extreme optical parameters, the relatively high level of residual aberrations, and the interface between the image and two-eye vision. As such, biocular magnifiers of reasonable magnifying power ($> \times 4.5$) cannot easily be analysed in a meaningful way using conventional optical design software. A number of years ago, a unique computer program was written¹ that enabled the analysis of biocular magnifiers in the way in which they were used, that is with the spatial image being viewed by two small mobile apertures of nominally fixed separation. Utilising the power of modern PC computers, this program has been extended considerably so that it provides detailed graphical analysis of the visual aberrations appropriate to an extreme optical system usable by both eyes together. The reasoning behind the software and examples of the graphical analysis of biocular designs is given in the paper.

1. INTRODUCTION

Variations in terminology in two-eye visual optics have led to a certain amount of confusion over what is meant by a given term. The following represent the generally accepted definitions² for the different types of two-eye viewing optical systems (Figure1):

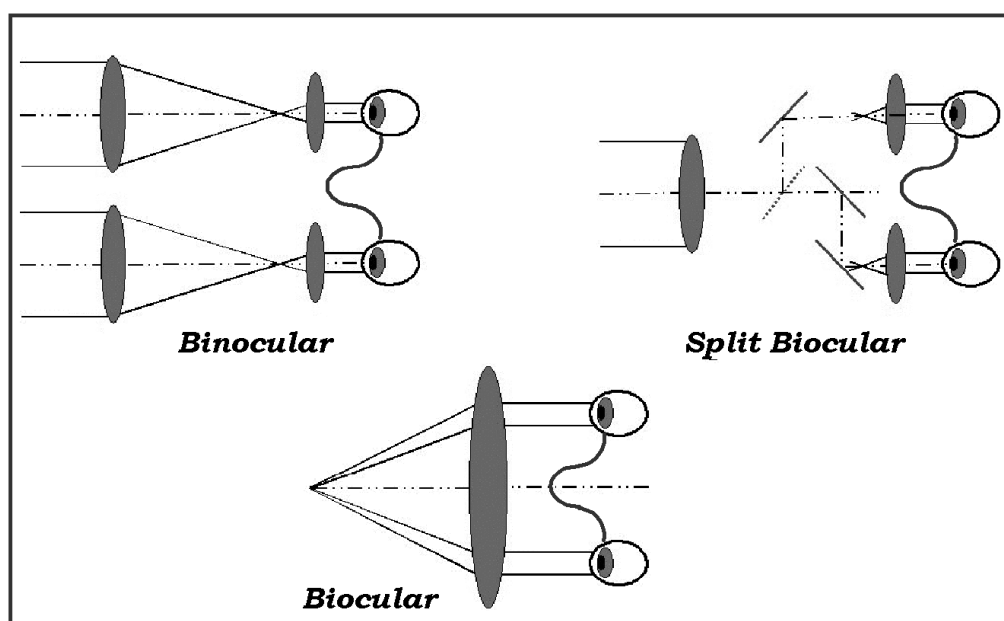


Figure 1: Two-eye optical system definitions²

Binocular: an optical system in which each eye is served by an independent optical train.

Split-biocular: an optical system having a common input axis in which some of the optics to each eye are separate.

Biocular: a single-axis optical system that is large enough for use by both eyes simultaneously.

The eyes are accustomed to operating as a pair and exhibit fusional lock, the ability to point accurately at the same part of a scene so that the brain receives images that are closely similar. When it is difficult to maintain this fusional lock, the observer experiences discomfort and fatigue. The ease with which fusion can be maintained when viewing through a biocular magnifier, plus the quality of the individual images and their similarity in size, is a measure of the success in the design. Any significant disparity between the distance at which the eyes accommodate (focus) when looking through the magnifier and that at which they can achieve fusional lock contradicts the eyes' natural muscular responses. Convergence of the optical axes of the eyes is natural, but divergence and dipvergence (different vertical tilt of the two axes) are not and cause discomfort.

2. HISTORY

Many years ago a novel computer program was described¹ that enabled the analysis of biocular magnifiers (a convenient term to describe non-collimated viewers with the eyes close-up to the optics). Until that time, conventional optical design software had generally treated these unusual systems as objective lenses, that is as though the integrated effect of the complete aperture should be considered rather than two small apertures of nominally fixed separation that are mobile within the entrance pupil. Unsurprisingly, many designers at that time concluded that biocular magnifiers of approaching x5 magnification or greater, typically requiring focal lengths <50mm, physical apertures >75mm and fields-of-view >40°, were impossible. The software, devised by Pilkington PE (now Thales Optics UK), enabled an optical designer to adjust the comparatively large residual aberrations to a level & balance that was acceptable to a pair of eyes operating together.

3. DESCRIPTION OF THE BIOCLULAR MAGNIFIER ANALYSIS PROGRAM

The biocular magnifier program operates by tracing a pair of rays from a point on the object (the display source), iterating on the direction cosines of these rays until they pass through the centre of each defined eye pupil. For a given object point, there will be a position in virtual image space where the two rays (lines-of-sight) approach each other most closely: at that position there will be a line-of-closest-approach. The locus of the centres of the lines-of-closest-approach, mapped over the 2-eye field of view, defines a Convergence Surface: that is, the surface over which the two eyes optical axes are most likely to converge (Fig 2).

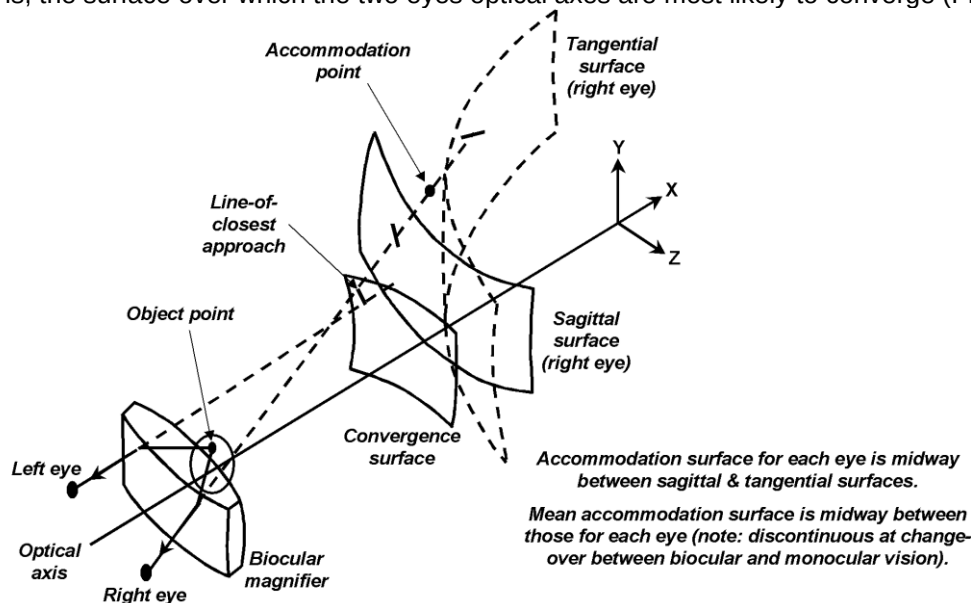


Figure 2: Biocular visual surfaces³

The eyes do not necessarily focus over the Convergence Surface. In fact there will be separate (pseudo-Coddington) sagittal (S) and tangential (T) foci along every line-of-sight for each eye. The locus of the mid-point between the S and T foci corresponding to every point on the image defines, for each eye separately, an Accommodation Surface: that is, the surface over which each eye is independently most likely to focus.

Using the surfaces defined in Fig 2, a number of important visual aberrations can be identified⁴ that together can be used to evaluate the performance of a biocular magnifier. These aberrations can be defined as:

Astigmatic difference: Separation between S & T surface (either eye), +ve if S focus nearer the observer.

Image curvature: Departure of the mean accommodation surface from a plane through its intersection with the optical axis, +ve if edge of the accommodation surface is further away from the observer than centre.

Binocular disparity (also called biocular defocus): Separation between the convergence and accommodation surfaces (either eye), +ve if accommodation is nearer the observer than convergence.

Dipvergence (also called supravergence): Vertical component of the line-of-closest-approach; represents the amount of vertical misalignment required of the optical axes of the two eyes in order to fuse the image.

Image distortion: Horizontal, vertical and modulus components defined by the program as the mean of those given by the two eyes.

Image resolution: Calculated in the program by a close ray trace in orthogonal planes around each line-of-sight, and given as a limiting resolution for each eye in sagittal & tangential orientations separately.

The coherent link between biocular magnifiers and the eye, and the non-linear behaviour of the latter, means that there is no direct analytical method for calculating tolerances for visual systems; these must instead be estimated from empirical data. The maximum values suggested in Figure 3 are based on extensive trials with a number of observers; in most cases the values are in reasonable agreement with those obtained by other authorities (for example^{5,6,7,8}), although some of these were investigating monocular optics.

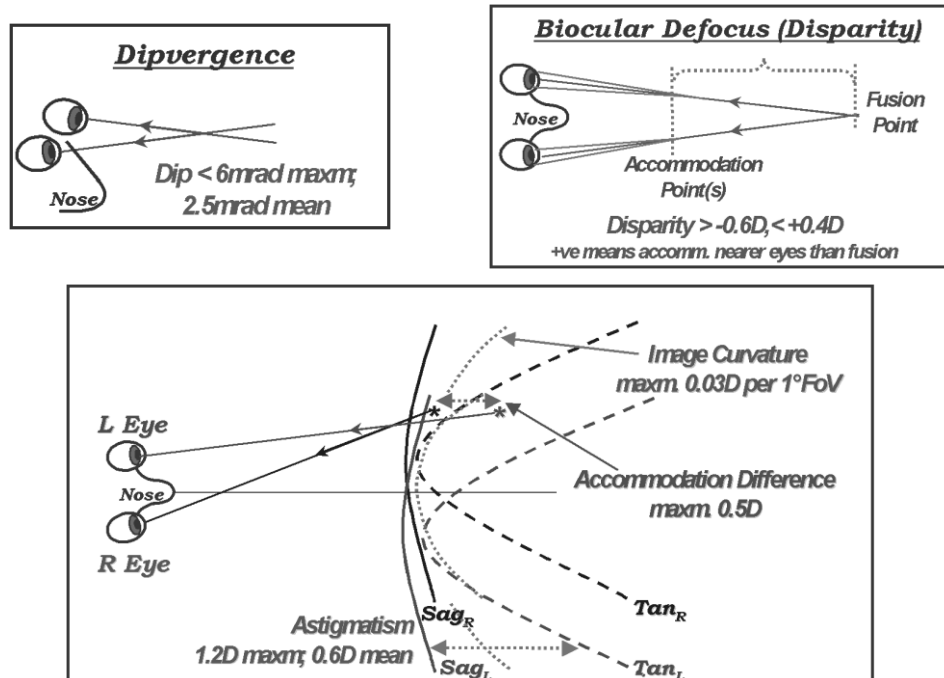


Figure 3: Tolerances on some important visual aberrations⁴

Figure 3 contains no tolerances on image distortion and resolution, this is because the extent to which the former is critical depends very much on the application: resolution should generally be eye-limited for at least one eye (or, at the worst, equivalent to the resolution of the display source).

EYE PAIR 1: LEFT EYE (Y = 0.0mm, Z = -32.0mm) RIGHT EYE (Y = 0.0mm, Z = +32.0mm)									
Object Point 60: Y -12.30mm Z 3.95mm X 0.00mm									
Accommodation Distance: 677.612mm Fusion Point: x 688.534mm y -132.724mm z 42.832mm									
Biocular Defocus: 0.023mrad Accommodation Difference 0.008D Astigmatism: L eye -0.073D R eye -0.225D									
Dipvergence 1.617mrad Image curvature 0.135D Resolution s/t: L eye 86/68 lp/mm R eye 23/26 lp/mm									
Direction Cosines		Sagittal Focus		Tangential Focus		Line of Closest Approach			
Left	Right	Left	Right	Left	Right	Mid Point	Left End	Right End	
*l 0.976235	0.981953	*x 692.722	735.742	*x 659.481	631.317	*x 688.534	688.534	688.534 *	
*m -0.188972	-0.188491	*y -134.092	-141.230	*y -127.657	-121.185	*y -132.724	-133.281	-132.168 *	
*n 0.106086	0.015461	*z 43.277	43.585	*z 39.665	41.940	*z 42.832	42.822	42.841 *	
Distortion: horizontal 1.0% vertical 2.7% modulus 2.9%									

Fig.4: Biocular magnifier analysis program – data output for just 1 eye position & 1 object point (expanded)
[Key: D = dioptres; mrad = milliradians; lp/mm = line-pairs per millimetre; all linear dimensions in millimetres]

Figure 4 shows the data (expanded & 'cleaned-up' for clarity) taken from the program for one position of the eyes and one object point, in general a large number of object points (>80) are mapped for more than one eye(s) position. The result, generated for several eye(s) positions, is a large amount of data requiring lengthy evaluation, even by an experienced user. It was considered, therefore, that the program could be improved by the addition of a graphical output that should make the plethora of data much easier to appreciate.

4. GRAPHICAL EXTENSION TO THE PROGRAM

The most difficult part in devising a graphical extension to a program that generates so much information for each point on the object is in deciding which information to display and in what form. It was considered that a diagram superficially similar to that of figure 2 would be appropriate, except that some of the information would have to be simplified. The principal simplification was to show only one accommodation surface, which would be the mean (where appropriate) of the surfaces for the individual eyes: this would result in a discontinuity in the surface at the edge of the biocular overlap region. Astigmatism in dioptres, given as a mean of the absolute values for the two eyes, would be shown as (coloured or greyscale) contours superimposed on the mean accommodation surface.

The convergence surface would be as in Figure 2 with the addition of contours representing the absolute value of dipvergence in milliradians. Distortion would be shown as a distorted grid, with the value in the biocular overlap region being the mean for the two eyes. Resolution in cycles/millimetre at the object would be given as the mean of sagittal and tangential values for monocular or biocular vision as appropriate, calculated assuming that the eyes are focused on the accommodation surface. The resolution would then be indicated as (coloured or greyscale) contours superimposed onto the distorted grid. Biocular disparity could be judged from the dioptric separation between accommodation & convergence surfaces.

5. BIOCLAR DESIGNS FOR ANALYSIS

Two biocular designs by TOUK (Thales Optics UK) that have already been described in the literature⁴, were chosen as being representative of two extremes in high power biocular magnifier design. The first – Figure 5a – is a straight design having a magnification near the lower limit of the category, but having good correction of image curvature, distortion and lateral chromatic aberration. The second – Figure 5b – is a design that is limited in the number of possible lens elements by the high magnification (and hence short focal length) and the internal space required by a prismatic fold. The relative simplicity of this design implies an 'on-the-limit' situation with regard to correction of all relevant aberrations.

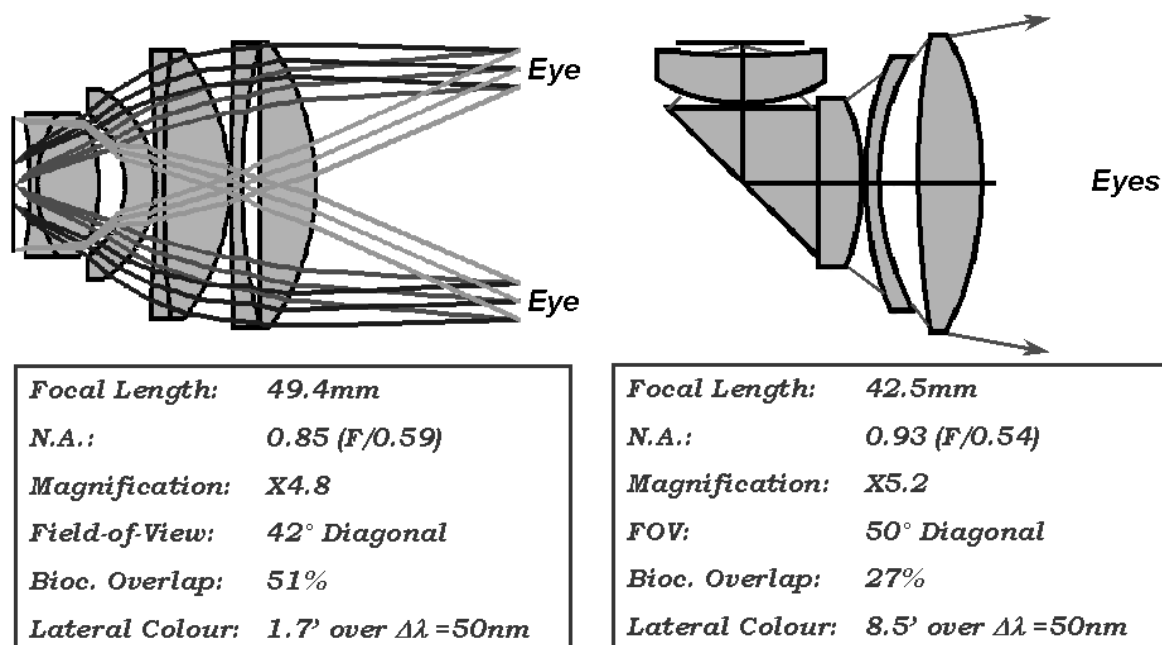


Figure 5a: Straight biocular design

Figure 5b: Folded biocular design

In its essential configuration the design in Figure 5a resembles an Amici microscope objective, hardly surprising given that a high numerical aperture is a prerequisite, although here there is also the requirement for a relatively large field-of-view ($>40^\circ$). Both separated and cemented doublet components are employed in the design in order to provide high correction of lateral colour. The strongly meniscus negative element fulfils two important functions: providing correction of the pincushion distortion that is inherent in the rest of the design, and also assisting in reducing the Petzval sum and hence curvature of the image.

The design in Figure 5b is a 'reductio ad absurdum' configuration containing the absolute minimum that is necessary to correct the most important aberrations. Spherical aberration, of several orders, is corrected by the weak but extreme meniscus negative lens near the front, given that a correct choice is made for the glass types of both it and the front lens element. Astigmatism is corrected by the convex surface of the lens element nearest the object. The relatively low level of correction of chromatic aberration is achieved simply by the use of a highly dispersive flint glass as the material of the fold prism.

Subsequent to the time at which the basic configuration of the biocular magnifier in Figure 5b was devised¹, one of the present authors claimed two new 'highs' for the design⁹: the highest-ever magnification for a folded biocular and a new high in the size of the residual aberrations! This highlighted again the absolute necessity of understanding how these residual aberrations should be balanced and analysed. It should also be noted that this folded biocular has been made and used very successfully in many thousands of units.

6. GRAPHICAL ANALYSIS

6.1 Introduction

The following figures are in three parts for each magnifier and different position of the eyes. The first part represents a 3-dimensional view of the accommodation & convergence surfaces, complete with astigmatism & dipvergence contours respectively, as seen from the eye(s). The second part is the same as the first but viewed from the opposite direction in that the surfaces may 'cut-through' each other. The third part is a 2-dimensional diagram showing a rectilinear grid as seen distorted by the biocular magnifier, with a potential discontinuity at the edge of the biocular overlap: resolution is shown as a superimposed contour map. Note that, in the case of astigmatic and dipvergence values, the colour or greyscale contours appear redder or lighter respectively when they approach the maximum values suggested in Figure 3.

6.2 High-Performance, Straight Biocular – Centred Eyes

Figures 6a, 6b and 6c show the visual performance for the straight biocular of Figure 5a, all figures being symmetrical in each quadrant due to the centred eyes. The relative complexity of the optical design and the moderate field-of-view mean that all of the visual aberrations do not approach the maximum recommended values. Note particularly the flat image and slightly overcorrected distortion, both of which are largely the result of employing an extreme meniscus component as the second from the left in Figure 5a. Dots in the X-axis represent increments of 0.1 dioptres and problem areas for astigmatism and dipvergence are indicated by sudden shade changes in the greyscale contours (much clearer in colour, contact the authors if interested in colour pictures).

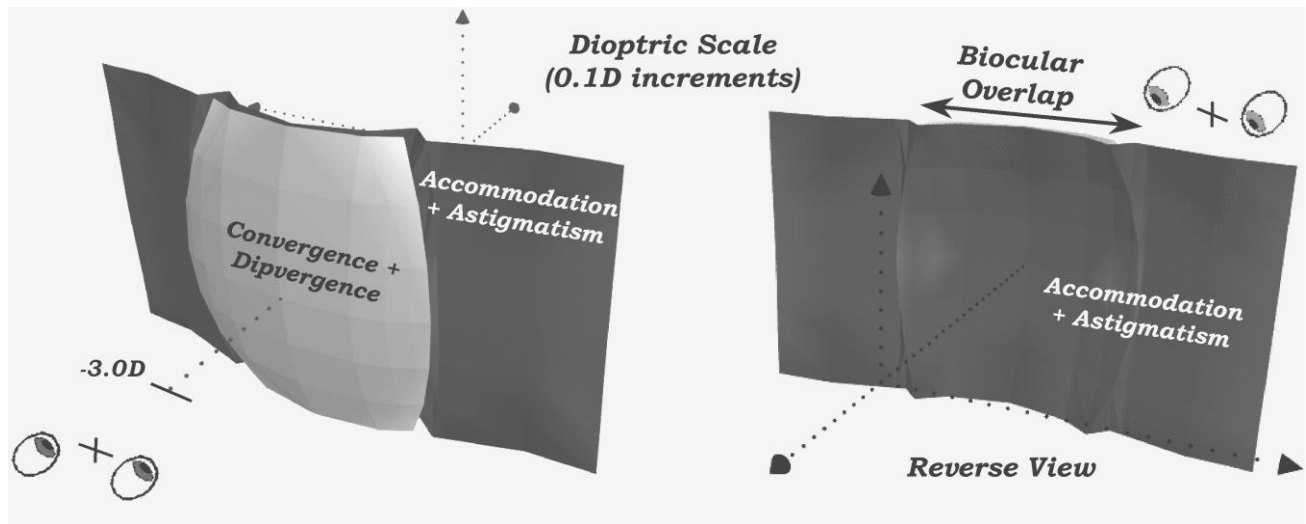


Figure 6a & 6b: Accommodation and convergence surfaces with superimposed astigmatism & dipvergence

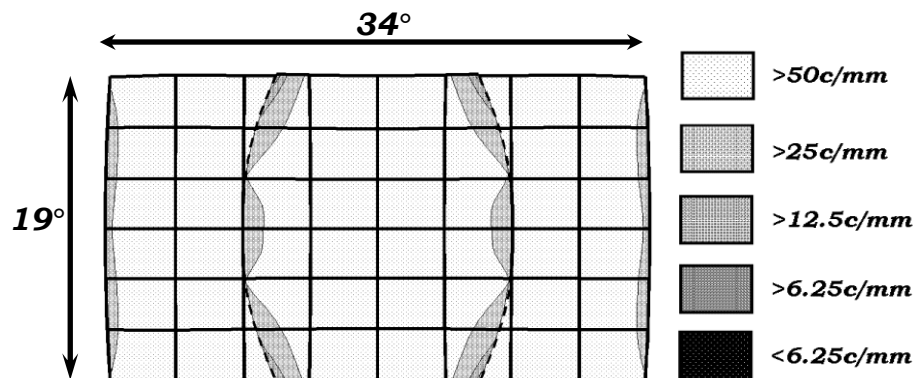


Figure 6c: Distorted grid with superimposed resolution (at the display object).
(Note the discontinuities at edges of biocular overlap)

6.3 High-Performance, Straight Biocular – Diagonally Shifted Eyes

Figures 7a, 7b and 7c show the visual performance for the straight biocular with eyes shifted 12mm up and 12mm to the right: here there is no symmetry because of the skewed position of the eyes. Visual performance is still good, the complete accommodation surface being within an envelope of less than 0.5dioptries and astigmatism rarely being more than 0.35dioptries (although it hits a peak of nearly 0.6dioptries in one extreme corner). The convergence surface has shifted to the left and has tilted by about 0.4dioptries with the left edge being further away and the right edge nearer relative to the observer.

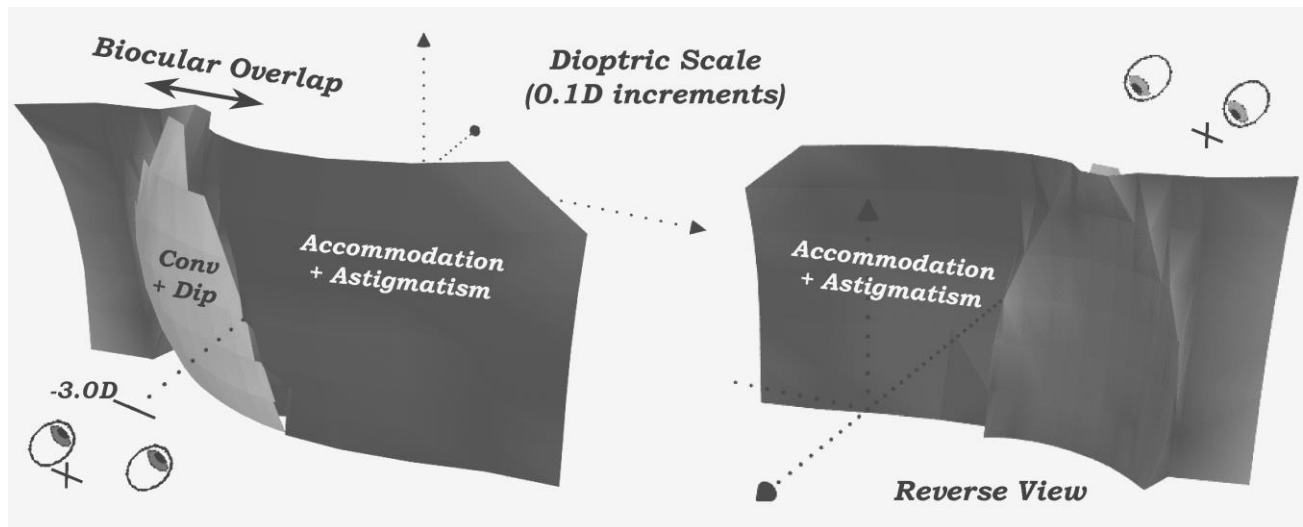


Fig. 7a & 7b: Accommodation and convergence surfaces with superimposed astigmatism & dipvergence

Distortion correction is still good and, while resolution is not quite as good as in the case of centred eyes, it is still quite good. As a guide, eye resolution for a 5-millimetre eye pupil is approximately 30c/mm in high brightness and contrast conditions, but lower in less ideal conditions.

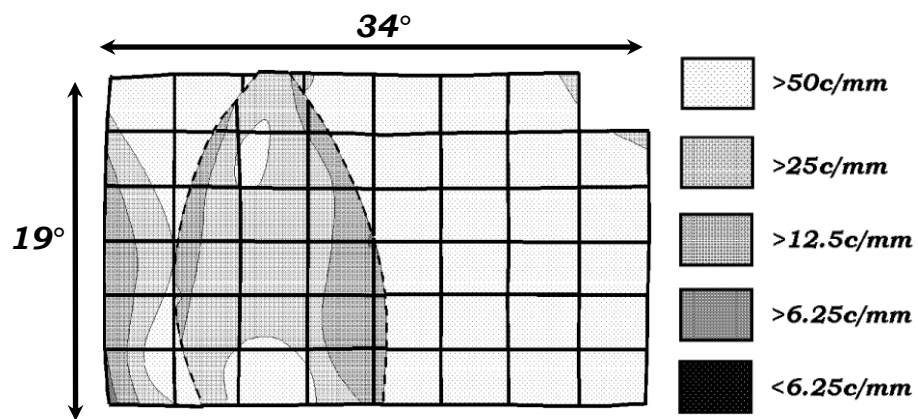


Figure 7c: Distorted grid with superimposed resolution (at the display object).
(Note the discontinuities at edges of biocular overlap)

6.4 On-the-limit Folded Biocular – Centred Eyes

Figures 8a, 8b and 8c show the visual performance for the folded biocular of Figure 5b, all figures being symmetrical in each quadrant due to the centred eyes. The higher level of visual aberrations by comparison with the straight-through biocular design are obvious, although relative simplicity of the design and the wider field-of-view make this inevitable. The level of field curvature is particularly noticeable, the dioptric difference between centre and extreme edge of the image being about 1.5 dioptres. Astigmatism is reasonable and indeed could be higher in order to provide an apparently flatter image (see, however, Section 6.5).

The convergence surface 'cuts through' the accommodation surface, although this is not necessarily a bad thing in that it means that, for most of the field-of-view, the binocular disparity is fairly well under control. Divergence values are very close to the limit recommended in Figure 3 (shown as lighter colours on the convergence surface – much clearer in a colour diagram)

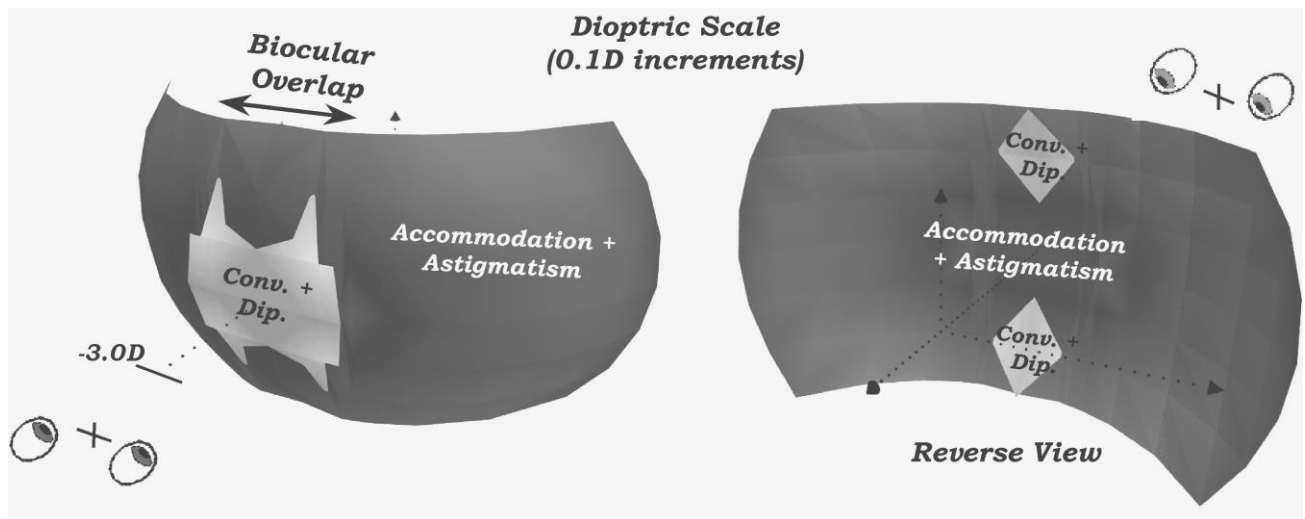


Figure 8a & 8b: Accommodation and convergence surfaces with superimposed astigmatism & dipvergence

The image distortion given by the design is different between monocular & biocular parts of the field-of-view, and between the horizontal & vertical directions in the monocular part. The latter gives rise to the characteristic image shape given by an on-the-limit biocular, often described as the 'side of a barrel' effect.

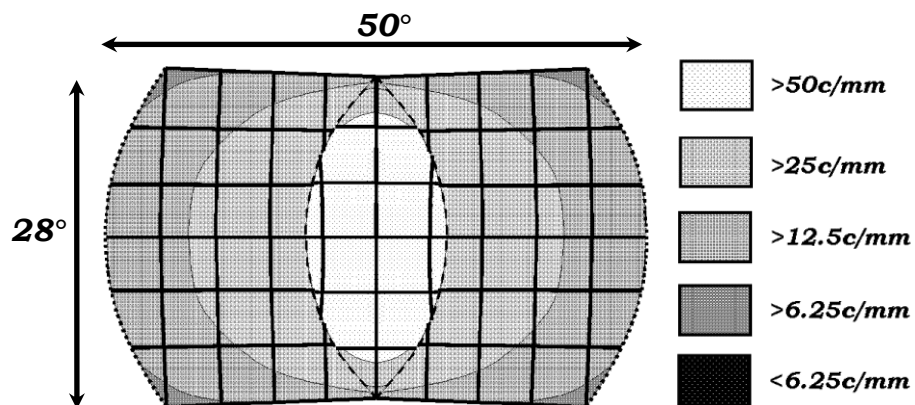


Figure 8c: Distorted grid with superimposed resolution (at the display object).
(Note the discontinuities at edges of biocular overlap)

6.5 On-the-limit Folded Biocular – Diagonally Shifted Eyes

Figures 9a, 9b and 9c show the visual performance for the folded biocular with eyes shifted 12mm up and 12mm to the right: here there is no symmetry because of the skewed position of the eyes. Now life starts to get interesting! All of the figures are complex and show a high level of visual aberration, although it should be pointed out that it was never intended that this biocular should be used from such an extreme position (there is a head-locating pad built into the system). Nonetheless it is interesting to see the form of the aberrations. Note also that the upper part of the image cannot be seen from this diagonal position.

One edge of the accommodation surface is at infinity while the other has remained in much the same position as that of the centred-eyes case: dioptric differences between centre and the two edges are about 2.3 and 1.5 respectively. Astigmatism reaches a value of about 1.2 dioptres at the 'bad' edge, thus inferring that one of the orientations is imaged well beyond infinity. The convergence surface is, however, quite well behaved, as is binocular disparity because the relevant (central) part of the accommodation surface is relatively flat. Dipvergence is, however, very high at the lower left corner of the biocular overlap.

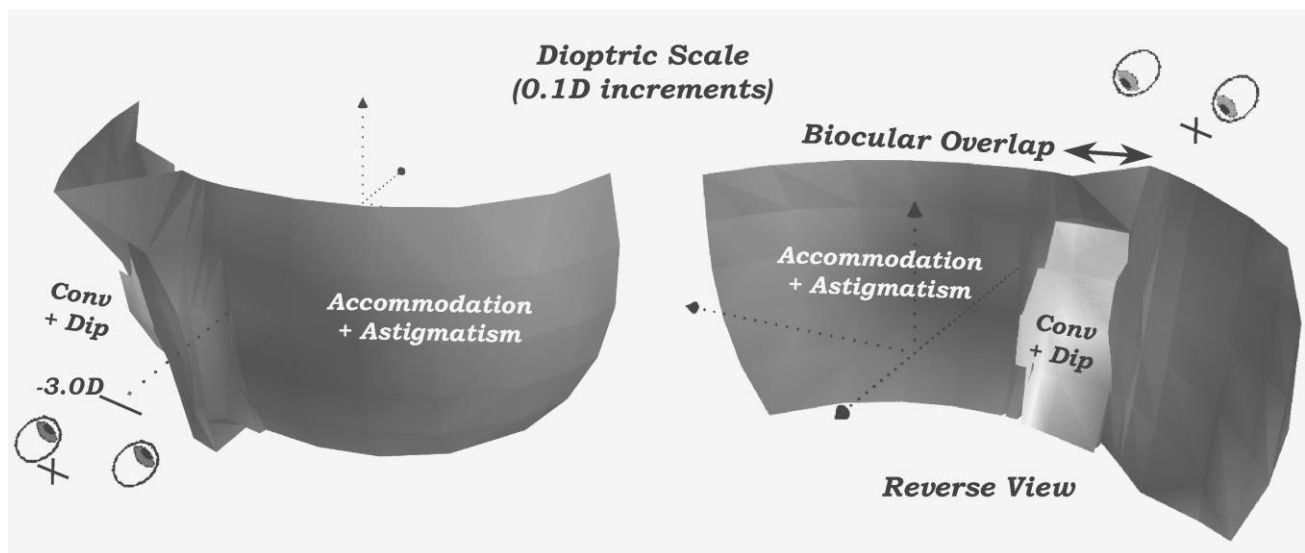


Figure 9a & 9b: Accommodation and convergence surfaces with superimposed astigmatism & dipvergence

Distortion is not that much different from the centred-eyes case, but resolution almost ceases to exist at the very extreme upper right and lower left parts of the whole image.

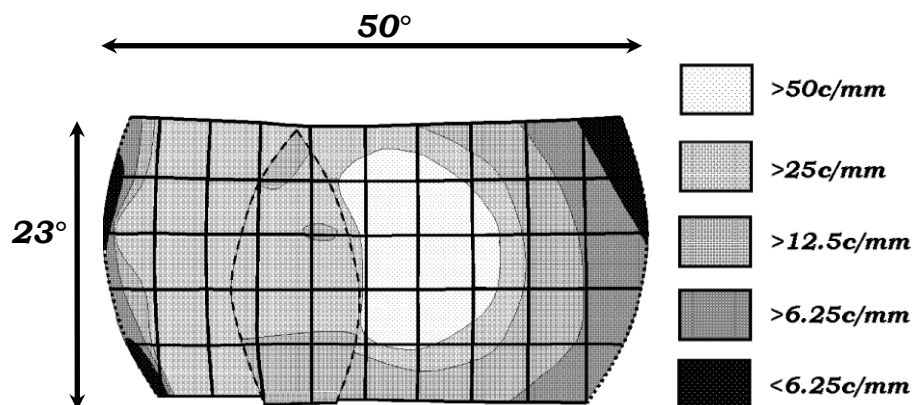


Figure 9c: Distorted grid with superimposed resolution (at the display object).
(Note the discontinuities at edges of biocular overlap)

7. SUMMARY

The Thales Optics UK biocular analysis software has been considerably enhanced by the addition of graphical output. Prior to this addition, it was necessary to scan through many pages of dense-packed data. The graphical enhancement has, therefore, considerably simplified the process of biocular magnifier design optimisation and analysis. A particular benefit in the new graphical output is that it is now extremely easy to pick out positions in the field of view where aberration limits are being approached or exceeded. These aberrations show up as curvatures or separations of the two visual surfaces, and (in the colour version of the program) warning colour contours superimposed on these surfaces. Whether the current form of graphical output is the optimum is open to argument, but it does represent a major advance and demonstrates a number of visual aberrations in an easy-to-assimilate form.

8. ACKNOWLEDGEMENTS

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9. REFERENCES

1. P. J. Rogers, "*Monocular and biocular magnifiers for night vision equipment*", pp.37-43, Proceeding of Electro-Optics International (Brighton, UK), 1972.
2. M. H. Freeman, "*On the use of the term 'biocular'*", p.266, Optics & Laser Technology Vol.5, 1973.
3. M. H. Freeman, "*Visual performance with biocular magnifiers*", p.266, Optics and Laser Technology Vol.7, 1975.
4. P. J. Rogers, M. H. Freeman, "*Biocular Display Optics*", Chapter 10 in Electro Optical Displays (ed. M. A. Karim), Marcel Dekker Inc., New York 1992.
5. W. N. Charman, H. Whitefoot, "*Astigmatism, accommodation, and visual instrumentation*", p.3903, Applied Optics Vol.17, 1978.
6. G. J. Burton, R. Home, "*Binocular summation, fusion and tolerances*", p.809, Optica Acta Vol.27, 1980.
7. M. K. Giles, "*Aberration tolerances for visual optical systems*", p.634, Journal of the Optical Society of America Vol.67, 1977.
8. G. J. Burton, N. D. Haig, "*Effects of Seidel aberrations on visual target discrimination*", p.373, Journal of the Optical Society of America Vol.1A, 1984
9. P. J. Rogers, "*Biocular magnifiers – a review*", pp.362-370, Proceedings of the International Lens design Conference (Cherry Hill, USA), SPIE Vol.554, 1985.